

User Requirements Specification (URS)

AI-native Quality & Supplier-Audit Platform for regulated supply chains — covering the foundational requirements and the next-generation "white-space" requirements.

Document type	User Requirements Specification (URS)
System	AI-native EQMS + Supplier/Vendor-Audit Management Platform
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Status	Draft for review — requirements to be ratified by Quality, IT/CSV, and Business owners
Regulatory context	FDA 21 CFR Part 11 & Parts 210/211/820; EU GMP Annex 11; ISO 9001 / 13485 / 14971 / 42001; ICH Q9 / Q10; FDA QMSR (2026)
Intended readers	Quality (URS owners), IT / CSV / validation, Product & Engineering, Procurement/Supply-chain

1. Purpose

This URS defines, in the users' terms, what the system must do — independent of how it is implemented. It is the basis for design specification (FS/DS), configuration, verification testing (IQ/OQ/PQ), and traceability. It is organised in two layers: **Part A — Foundational requirements** (the established capabilities a regulated EQMS + supplier-audit platform must provide) and **Part B — White-space requirements** (the next-generation capabilities that address the unmet, structural problems of the category).

2. Scope

- In scope:** internal quality management (the EQMS module set); supplier/vendor qualification and audit management; the cross-cutting platform services (security, e-signature, audit trail, AI governance, configuration, integration, reporting).
- Out of scope (this URS):** physical laboratory instrumentation, ERP/MES core functionality (interfaced, not replaced), and hardware procurement. These are referenced only where the system must integrate with them.

3. Definitions & conventions

Requirement ID format: URS - [AREA] - [nnn]. **Priority (MoSCoW):** M Must S Should C Could w Won't-yet (future). Each requirement is testable and traceable. ALCOA+ = Attributable, Legible, Contemporaneous, Original, Accurate (+ Complete, Consistent, Enduring, Available).

Area codes: GEN (general/platform), SEC (security/access), ESG (e-signature), ATR (audit trail), DOC, DEV (deviation), CAP (CAPA), CHG (change control), TRN (training), CMP (complaint/event), RSK (risk), MRV (management review), SUP (supplier/audit), CFG (configuration), INT (integration), RPT (reporting), AIG (AI governance), and the white-space areas WS1-WS6.

How to use this URS. Part A requirements are the regulated baseline — most map to capabilities standard in the category. Part B requirements are the differentiating, forward-looking set derived from the documented white spaces of the domain. Priorities are indicative for a draft; ratify them with the business before this URS drives design and validation. In a regulated programme this URS is a controlled document.

A1 · Platform, security, records & signatures

Priority: M Must · S Should · C Could

General & platform (GEN)

ID	Requirement — the system shall...	Pri	Anchor
URS-GEN-001	Be a multi-tenant, cloud-deliverable platform, with each tenant's data logically isolated.	M	Annex 11
URS-GEN-002	Support cloud, hybrid, and on-premise deployment of the application without change to user-facing function.	S	—
URS-GEN-003	Provide role-based dashboards and a task inbox (assignments, approvals, due/overdue items) per user.	M	ICH Q10
URS-GEN-004	Operate in a validated state; provide documentation supporting IQ/OQ/PQ and clearly delineate vendor-vs-customer validation responsibility.	M	GAMP 5
URS-GEN-005	Be available with a defined uptime target and support disaster recovery (RPO/RTO) appropriate to GxP records.	S	Annex 11

Security & access control (SEC)

ID	Requirement — the system shall...	Pri	Anchor
URS-SEC-001	Enforce role-based access control (RBAC) with configurable roles and per-capability permissions.	M	Part 11
URS-SEC-002	Support individual named accounts only; prohibit shared logins for any action that can be signed.	M	Part 11
URS-SEC-003	Support SSO and enforce multi-factor authentication; configurable session timeout and logout.	M	Annex 11
URS-SEC-004	Enforce separation of duties (e.g. system administrator ≠ quality approver) and make any role-overlap explicit and auditable.	M	Part 11
URS-SEC-005	Constrain role scope by site, supplier, or module where required.	S	—

Electronic signature (ESG)

ID	Requirement — the system shall...	Pri	Anchor
URS-ESG-001	Apply 21 CFR Part 11 / Annex 11-compliant electronic signatures binding signer identity, meaning, timestamp, and the exact record version.	M	Part 11
URS-ESG-002	Require re-authentication (credentials/PIN/MFA) at the moment of signing.	M	Part 11
URS-ESG-003	Maintain a configurable list of signature meanings (e.g. Authored, Reviewed, Approved).	M	Part 11
URS-ESG-004	Lock a signed record against edit; require a new controlled version for any change.	M	Part 11

Audit trail & data integrity (ATR)

ID	Requirement — the system shall...	Pri	Anchor
URS-ATR-001	Maintain a secure, time-stamped, append-only audit trail of all create/modify/delete/transition/sign actions, capturing actor, timestamp, before/after state, and reason-for-change.	M	Part 11 / ALCOA+
URS-ATR-002	Make the audit trail human-readable, searchable, and exportable, and prevent its modification by any user.	M	Annex 11
URS-ATR-003	Ensure all records meet ALCOA+ and enforce a configurable data-retention and legal-hold policy.	M	ALCOA+
URS-ATR-004	Provide tamper-evident record integrity (e.g. per-record cryptographic hashing of signed snapshots).	S	ALCOA+

A2 · EQMS process modules

The system shall provide the canonical, interconnected EQMS modules, each with a configurable workflow/state machine, mandatory-field rules, e-signature gates, and full audit-trail capture.

Document control (DOC)

ID	Requirement — the system shall...	Pri
URS-DOC-001	Manage the controlled-document lifecycle: author → review → approve → effective → periodic review → retire, with enforced version control and a single current effective version.	M
URS-DOC-002	Retain full version history with the ability to view superseded versions and the change summary between versions.	M
URS-DOC-003	Trigger downstream actions on document change (e.g. assign retraining to affected roles).	S

Deviation / non-conformance (DEV)

ID	Requirement — the system shall...	Pri
URS-DEV-001	Capture a deviation/NC with structured intake, risk classification, and the affected product/process/batch.	M
URS-DEV-002	Support a structured investigation (incl. root-cause method such as 5-Why) and a documented disposition with product-quality/patient-safety impact assessment.	M
URS-DEV-003	Link a deviation to one or more CAPAs and propagate the linkage end-to-end.	M

CAPA (CAP)

ID	Requirement — the system shall...	Pri
URS-CAP-001	Manage the closed-loop CAPA lifecycle: initiate → investigate root cause → action plan → implement → verify effectiveness → close.	M
URS-CAP-002	Prevent closure until effectiveness verification is recorded; support reopen/supersede.	M
URS-CAP-003	Enforce risk-based due dates and escalate overdue actions.	S

Change control (CHG)

ID	Requirement — the system shall...	Pri
URS-CHG-001	Manage change requests with impact/risk assessment, multi-step (sequential or parallel) approval, and controlled implementation.	M
URS-CHG-002	Link a change to affected documents, validation, training, and (where relevant) supplier re-qualification.	M

Training, complaint, risk, management review (TRN / CMP / RSK / MRV)

ID	Requirement — the system shall...	Pri
URS-TRN-001	Assign, track, and record training and competency against roles/SOPs; auto-assign on document revision.	M
URS-CMP-001	Manage complaints/events: intake → triage → investigation → regulatory-reporting assessment → CAPA linkage.	M
URS-RSK-001	Provide quality risk management (identify → assess → control → review) usable standalone and embedded in other modules (ICH Q9).	M
URS-MRV-001	Aggregate cross-module quality KPIs into a management-review view supporting periodic governance (ICH Q10).	S

Interconnection requirement (the digital thread). URS-GEN-006 M — The system shall maintain end-to-end traceability across modules so that a record in one process (e.g. a deviation) can create, link to, and be traced from a record in another (e.g. a CAPA, a change, a for-cause audit, a training assignment), with the full chain visible and audit-trailed.

A3 · Supplier / vendor-audit management

The system shall manage the full supplier qualification-and-audit lifecycle as a process distinct from, but linked to, internal quality.

Supplier qualification & risk (SUP — qualification)

ID	Requirement — the system shall...	Pri
URS-SUP-001	Maintain a supplier master with qualification status, approved-supplier-list membership, certifications, and material/service criticality.	M
URS-SUP-002	Support initial supplier evaluation via questionnaire and document review, and risk-classify suppliers to set audit depth and cadence.	M
URS-SUP-003	Capture quality agreements and link them to the supplier record.	S
URS-SUP-004	Continuously monitor supplier performance (defects, complaints, delivery) and re-score risk on a defined cadence.	S

Audit lifecycle (SUP — audit)

ID	Requirement — the system shall...	Pri
URS-SUP-010	Manage the audit lifecycle (plan → prepare → conduct → findings → report → sign-off → close) with configurable phases and milestones, and support supplier, internal, and for-cause audit types.	M
URS-SUP-011	Support pre-audit preparation: scope, auditor qualification/assignment, agenda, pre-audit questionnaire, pre-read collection, risk review.	M
URS-SUP-012	Support on-site/remote execution: questionnaire-driven evidence capture, observation logging, and finding classification by criticality.	M
URS-SUP-013	Support post-audit follow-up: cited audit report, supplier CAPA request, CAPA evaluation, and the approval decision — all e-signed and audit-trailed.	M
URS-SUP-014	Maintain a searchable, reusable repository of completed audits, exportable as controlled, versioned reports.	S
URS-SUP-015	Trigger a for-cause audit automatically from defined events (recurring deviation, recall, regulatory action, failed CAPA).	S

Configuration, integration, reporting (CFG / INT / RPT)

ID	Requirement — the system shall...	Pri
URS-CFG-001	Allow administrators to enable/disable modules, configure workflows/states/transitions, mandatory fields, and approval steps without code change.	M
URS-CFG-002	Maintain a configurable standards/controls library (e.g. GMP, 21 CFR, ISO) as data, and per-tenant vocabulary overrides.	S
URS-INT-001	Integrate with ERP/LIMS/MES and a document repository via documented APIs; avoid duplicate manual data entry.	M
URS-RPT-001	Provide configurable dashboards, trend analysis, and exportable reports across modules and sites.	M

A4 · AI governance (baseline)

Where the system uses AI/ML, it shall do so under explicit governance. These baseline AI requirements apply to *any* AI capability; the differentiating AI requirements are in Part B (WS2).

ID	Requirement — the system shall...	Pri
URS-AIG-001	Treat AI as decision-support only: AI may draft, suggest, or analyse, but shall never commit, approve, or sign a record — a qualified human always reviews and signs.	M
URS-AIG-002	Allow each AI capability to be enabled/disabled per tenant and per module/process.	M
URS-AIG-003	Continue to function as a deterministic workflow/records system with all AI disabled (AI is additive, not a dependency).	S
URS-AIG-004	Redact or protect PII/sensitive data before it is sent to any external model.	M
URS-AIG-005	Govern AI access by the same RBAC as the rest of the system (only permitted roles may invoke AI).	M
URS-AIG-006	Where AI is used in GxP-impacting steps, support an AI-governance framework consistent with ISO 42001 and the organisation's validation approach.	S

The bridge to Part B. Part A above is the regulated baseline — broadly the capability set a mature EQMS + supplier-audit platform is expected to provide. The requirements that *differentiate* the system, and that address the documented structural gaps of the category, follow in Part B. They are written to the same testable standard, with priorities reflecting that some are foundational-differentiators (Must/Should) and some are forward bets (Could/Won't-yet).

Traceability note. Every requirement ID here is intended to trace forward to a design specification item and to one or more verification tests (OQ/PQ), and backward to a business/regulatory driver. A traceability matrix (URS → FS/DS → Test) should be maintained as a companion artefact and is out of scope for this draft.

B1 · The differentiating requirements (1 of 3)

Part B translates the six documented white spaces of the domain into testable requirements. Each white space is introduced, then specified.

WS1 — Couple internal quality to a live supplier-audit network. The deepest structural gap: internal-quality systems and supplier-audit networks evolved separately and don't converge. These requirements close the seam at the company boundary.

ID	Requirement — the system shall...	Pri
URS-WS1-001	Treat a supplier-audit finding as a first-class input to internal quality: a finding shall be able to create a linked CAPA/change and be traced both ways.	M
URS-WS1-002	Allow a completed audit to be shared with, or reused by, multiple authorised buying organisations under controlled permissions (shared-audit model), without re-running the audit.	C
URS-WS1-003	Maintain a supplier identity that is consistent across audits and buyers (entity resolution), so risk and history aggregate per real-world supplier.	S
URS-WS1-004	Enable a buying organisation's qualified suppliers to become discoverable participants in an audit network, with consent and access controls.	W
URS-WS1-005	Govern multi-party data sharing with explicit, auditable consent and selective disclosure (who sees which fields of an audit/record).	S

WS2 — Reproducible, defensible AI in validated environments. Everyone has AI; few can prove, per decision, exactly how an output was produced in a form an inspector accepts. These requirements make AI defensible.

ID	Requirement — the system shall...	Pri
URS-WS2-001	Record, for every AI-assisted output, a reproducibility record capturing the inputs/sources retrieved, the prompt/version, the model and version, and a confidence indicator.	M
URS-WS2-002	Ground every AI output in retrievable source data and cite each material claim to its source; where grounding is insufficient, return an explicit "insufficient evidence" result rather than an unsupported answer.	M
URS-WS2-003	Make the AI reproducibility record part of the audit trail and presentable to an inspector on demand.	M
URS-WS2-004	Allow administrators to set a grounding/confidence threshold governing AI conservatism, and to choose/route the underlying model (incl. an on-premise/self-hosted model) per capability.	S
URS-WS2-005	Meter and log AI usage per action for cost transparency and governance.	C

B2 · The differentiating requirements (2 of 3)

WS3 — Fuse live external/public data into quality & supplier-risk decisions. A large stream of public regulatory signal exists but is rarely fused, as it changes, into quality decisions. These requirements close the gap between "what the regulator already knows" and "what my system reflects."

ID	Requirement — the system shall...	Pri
URS-WS3-001	Ingest public regulatory data (e.g. FDA inspections/warning letters/recalls, EMA, WHO) and associate it with the relevant supplier/site.	S
URS-WS3-002	Fuse that data into a continuously updated supplier-risk profile across multiple dimensions (regulatory history, recalls, responsiveness, trend, network exposure), with provenance tagging of every signal.	S
URS-WS3-003	Raise early-warning alerts when new external signals materially change a supplier's risk.	C
URS-WS3-004	Use external-data-derived risk to inform audit cadence and questionnaire focus (risk-weighted audit prep).	C

WS4 — True interoperability & the cross-system digital thread. Integration across the ERP/LIMS/MES/PLM/QMS landscape is the category's #1 failure mode. These requirements target an open, reliable digital thread beyond a single vendor's modules.

ID	Requirement — the system shall...	Pri
URS-WS4-001	Expose open, documented, versioned APIs for all major objects (events, CAPAs, audits, suppliers, documents) supporting both read and write.	M
URS-WS4-002	Support event-driven integration (webhooks/messaging) so external systems can react to quality events in near-real-time.	S
URS-WS4-003	Maintain provenance and a common data model across integrated sources so a quality event is traceable across system boundaries (the cross-system digital thread).	S
URS-WS4-004	Provide pre-built connectors for common ERP/LIMS/MES and document systems, and a documented pattern for custom connectors.	C
URS-WS4-005	Handle unstructured inbound data (scanned reports, certificates, emails) via document intelligence/OCR, extracting structured fields with provenance to the source.	S

B3 · The differentiating requirements (3 of 3)

WS5 — Affordable, validated, intelligent quality for smaller / emerging-market manufacturers. The capability frontier and the affordability frontier don't coincide. These requirements put validated, AI-enabled quality within reach of a small or export-oriented SMB.

ID	Requirement — the system shall...	Pri
URS-WS5-001	Support near-zero-touch onboarding via pre-configured, standard-aligned templates so a small manufacturer can be productive without a long implementation.	S
URS-WS5-002	Provide a low/no-cost evaluation path (e.g. scoped proofs-of-concept on the customer's own data) before commitment.	C
URS-WS5-003	Offer transparent, usage-aligned commercial terms suitable for small teams, with read-only/viewer access at low or no cost.	C
URS-WS5-004	Be usable by a lean quality team — minimal training to perform a core task — without sacrificing compliance controls.	S

WS6 — A genuinely industry-agnostic compliance engine. The canonical processes are abstractly the same across regulated industries, yet software is usually single-regime. These requirements specify a configurable engine instantiable per industry.

ID	Requirement — the system shall...	Pri
URS-WS6-001	Separate the fixed processing engine from industry-specific configuration (standards, vocabulary, rules, workflows, report templates), all held as data.	S
URS-WS6-002	Allow a new industry/regulatory regime to be added as a configuration pack, without changing the core pipeline.	C
URS-WS6-003	Validate data against any configured standard's controls (rules + thresholds + required evidence) generically.	S
URS-WS6-004	Produce the appropriate validation/compliance/certification output for the configured industry from a common reporting engine.	C

4. Assumptions, constraints & acceptance

- Assumptions:** users are trained on their roles; the customer owns process SOPs; connectivity to integrated systems is available; the customer performs its portion of validation.
- Constraints:** the system must operate within applicable data-residency and privacy law; AI features are subject to the customer's validation and AI-governance policy; on-premise model deployment requires defined validation (IQ/OQ/PQ) before GxP use.
- Acceptance:** each requirement is accepted when a corresponding verification test passes and is traceable in the URS→FS/DS→Test matrix. Must-priority requirements are mandatory for release; Should/Could are scheduled by the programme.

Status & honesty note. This is a DRAFT URS for review. Priorities (MoSCoW) are indicative and must be ratified by Quality, IT/CSV, Procurement, and Business owners before driving design and validation. Part A reflects the regulated baseline for the category; Part B reflects the documented structural white spaces and is intentionally forward-looking — several Part B items are differentiators or future bets, not assertions that they are already built. Regulatory references are current to early 2026 and must be confirmed against primary sources. This URS is, in a regulated programme, itself a controlled document and must be versioned and approved.